Yousaf Rakhshan

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EDUCATION

Nottingham Trent University | 2023–2026

- BSc (Hons) Computer Science – Artificial Intelligence | Upper Second-Class Honours (2:1)
- Relevant modules: Machine Learning (Foundations and Advanced), Natural Language Processing, Cognitive Computing, Distributed Database Engineering, Ethical AI, Big Data Programming

PROJECTS AND EXPERIENCE

AI-Integrated iOS Birthday Social Media and Health Platform

- Technologies: Swift, Xcode, Core ML, Computer Vision, Firebase, AI APIs
- Architected a native iOS application using Core ML and computer vision for image training, birthday recognition and health diagnostics.
- Integrated Firebase to provide a real-time database architecture supporting concurrent user interactions and data persistence.
- Implemented cognitive computing features to support autonomous decisions based on environmental sensor inputs and personalised user engagement.

Commercial Full-Stack Web Platform | Pyala | October 2025–Present

- Website: www.pyala.co.uk
- Independently designed, developed and maintain a production-ready full-stack web solution for a local café.
- Implemented SEO-optimised front-end interfaces to improve online visibility and support customer conversion.
- Integrated the in-store till system and continually optimise website performance and reliability.

ACADEMIC COURSEWORK AND RESEARCH

- Developed supervised and unsupervised predictive machine learning models using large-scale industrial datasets, feature engineering and cross-validation to optimise model reliability and performance.
- Designed horizontally scalable distributed database architectures with fault-tolerance mechanisms for big data workloads.
- Applied cognitive computing principles to develop autonomous systems capable of responding to environmental inputs and making intelligent decisions.
- Evaluated ethical AI frameworks, focusing on bias mitigation, privacy by design (GDPR) and real-world societal impact.

TECHNICAL SKILLS

- AI and Machine Learning: Supervised and unsupervised machine learning, natural language processing, cognitive computing, autonomous decision systems, predictive modelling, ethical AI frameworks, bias mitigation
- Software Engineering: Full-stack web development, Swift and iOS development, Xcode, software design and implementation, systems software, API integration, Docker, Linux, Git, Agile/Scrum
- Data Engineering: Distributed database engineering, database management, data analytics, big data programming, real-time data synchronisation
- Programming: Python, C++, SQL, Swift (basic proficiency)
- Tools and Platforms: Microsoft Excel, Microsoft Word, Jupyter Notebook, TensorFlow, Visual Studio Code, Firebase, Kaggle, VMware, Ubuntu, Android Studio, Claude, Cursor, Codex, Kimi, Supabase

CERTIFICATIONS

- AI Compliance and Ethics for Data Centres — In progress

ADDITIONAL INFORMATION

- Interests: Skiing, travel, TED Talks, digital technology, robotics and football